COVALENT TRACTABILITY ASSESSMENT

KRAS — G12C

GTPase KRas (G12C mutant)

Indication: NSCLC, CRC, pancreatic cancer (KRAS G12C-mutant)
Target class: Gtpase | Modality: Covalent small molecule
Covalent confidence: ESTABLISHED

Premium Dataset — Product Summary

AXARA Biosciences | axara.bio

THE DECISION THIS ASSESSMENT ANSWERS

Should your team pursue a covalent strategy against **KRAS G12C**? And if yes — what warhead, what scaffold, what selectivity risks, and what resistance liabilities should you design around from day one?

Every covalent program starts with this question. Most teams spend 4–8 weeks answering it. This assessment delivers the answer immediately.

IF YOU DON'T BUY THIS ASSESSMENT

Your team will still need these answers. They will spend **40–80 hours** of comp chem time mining literature, running pocket analyses, building warhead models, and compiling selectivity data. That costs **$15,000–$25,000 in FTE time**, delays your first warhead decision by **2–4 weeks**, and produces an internal slide deck that lives in someone's Google Drive.

IF YOU BUY THIS ASSESSMENT

Your team starts with the answers. A computational chemist reviews the dataset in an afternoon. Your warhead selection meeting happens **this week**. Every deliverable is **machine-readable** — JSON, CSV, SDF integrate directly into your pipeline.

$2,500	2–4 wks	$15–25K	6–10×
COST	TIME SAVED	FTE AVOIDED	ROI

COST COMPARISON: IN-HOUSE vs AXARA

Task	In-House	AXARA
Pocket druggability analysis	8–16 hrs comp chem	**Included**
Warhead feasibility assessment	8–12 hrs	**Included**
Scaffold mining + seed ID	16–24 hrs med chem	**Included (curated)**
Resistance variant assessment	4–8 hrs	**Included**
Competitive landscape mapping	4–8 hrs BD/strategy	**Included**
Mass spec adduct tables	2–4 hrs proteomics	**Included**
Compile for team review	4–8 hrs	**Included (PDF + data)**
TOTAL	**$7K–$14K + 2–5 wks**	**$2,500**

COMMON OBJECTIONS

"We can do this ourselves."

Yes — in 2–6 weeks at $7K–$25K FTE cost. This assessment delivers the same answers in an afternoon, freeing your team to focus on compound design.

"It's computational, not experimental."

Correct. This is the computational triage that precedes experiments. It tells you which experiments to run, which warheads to prioritise, and which selectivity liabilities to watch.

"$2,500 needs approval."

The final page includes a ready-to-forward purchase justification with ROI calculation.

WHAT'S IN YOUR KRAS ASSESSMENT

Deliverable	Formats	What It Tells You
Covalent Tractability Map	PDF, CSV	Druggability verdict with confidence scores
Scored Pocket Structure	PDB	Loadable in PyMOL/Chimera with annotations
Pocket Druggability Analysis	JSON, HTML	fpocket volume, hydrophobicity, druggability
Mass Shift Table	JSON, HTML, CSV	LC-MS/MS adduct masses per warhead
Scaffold Library (curated)	JSON, HTML, CSV, SDF	Literature seeds + chemical biology probes
Resistance Variant Analysis	JSON, HTML	Clinical + predicted mutations scored
Competitive Landscape	JSON, HTML	Competitor pipeline by stage and modality
Bench Protocol	JSON, HTML	Construct, buffer, cell lines, controls, failure modes
Digital Twin	JSON	Machine-readable target model
Master Report	PDF	15–20 page narrative with decision framework
Validation Certificate	PDF	Engine version, provenance, timestamp

Total: 28+ files **| Formats:** CSV, HTML, JSON, PDB, PDF, SDF

All JSON files are API-ready. SDF files load directly into modelling software.

KRAS G12C — ASSESSMENT HIGHLIGHTS

Warheads evaluated: 4 covalent warhead classes

Preferred warhead: acrylamide

Scaffold seeds: 3 literature-validated

Competitors: 4 programs (2 approved)

Resistance variants: 4 (4 clinical)

Covalent confidence: ESTABLISHED

METHODOLOGY

• Pocket scoring: fpocket 4.0 on crystal structure (PDB: 6OIM)
• Scaffolds: Literature-mined with PMID provenance for every seed
• Resistance: Structural impact scoring on clinical + predicted mutations
• Engines versioned, reproducible, auditable

SAMPLE INSIGHT

"KRAS G12C is a clinically validated covalent target. The dominant clinical warhead is acrylamide. 4 resistance mutations reported in patients."

The full assessment contains the quantified scores, structural rationale, and actionable recommendations behind this summary.

PRICING

Single target	**$2,500**
5-target bundle	**$10,000 ($2,000 / target)**
Full panel (30+ targets)	**$45,000 ($1,500 / target)**

WHAT HAPPENS AFTER PURCHASE

✓ Secure download link within 1 business day
✓ All files immediately usable — no software licence required
✓ JSON/CSV/SDF integrate into any comp chem pipeline
✓ PDF master report ready for team circulation
✓ You own the dataset — no subscription, no expiry

PURCHASE JUSTIFICATION

Copy-paste into your internal approval request:

> "Requesting approval for a $2,500 covalent tractability assessment for KRAS from Axara Biosciences. Replaces 2–4 weeks and $15K–$25K of in-house computational triage. Includes curated scaffold library with literature provenance, resistance variant analysis, competitive landscape, mass spec adduct tables, and pocket druggability scoring. Machine-readable for pipeline integration. ROI: 6–10× vs in-house cost."

REQUEST ACCESS

hello@axara.bio | axara.bio/covalent

AXARA Biosciences — Computational drug discovery infrastructure — 2026